A 41-dimensional counterexample to convexity of Fisher information along the heat flow

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1 Problem and notation

For $a > 0$, write $\gamma_a(x) = (2\pi a)^{-d/2}\exp(-|x|^2/(2a))$ for the centered Gaussian density of variance a on $\mathbb{R}^d$. The Gaussian heat flow is $P_t f = f * \gamma_t$ for $t > 0$, with $P_0 f = f$; it satisfies $\partial_t P_t f = \Delta P_t f/2$. For a smooth positive probability density f, its Fisher information along this flow is

$$I_f(t) = \int_{\mathbb{R}^d} |\nabla \log(P_t f)|^2 P_t f \, dx. \tag{1.1}$$

The convexity question asks whether $I_f''(t) \geq 0$ for every such density whenever the derivatives exist. We present a proposed computer-assisted counterexample in dimension 41.

Theorem 1.1 (Failure of convexity). *The smooth positive probability density f defined in Section 2 satisfies $D(f) < -10^{-44}$, where D is defined below. There exists $\varepsilon > 0$ such that I_f is twice continuously differentiable and $I_f''(t) < 0$ for every $0 < t < \varepsilon$. Consequently, I_f is not convex on $(0, \infty)$.*

For a smooth positive function p, set

$$H_p = -\nabla^2 \log p, \qquad K_p = |\nabla H_p|_F^2 + 2\mathrm{tr}(H_p^3), \tag{1.2}$$

$$D(p) = \int_{\mathbb{R}^d} p K_p \, dx, \tag{1.3}$$

when the last integral is absolutely convergent. Here $|\nabla H|_F^2 = \sum_{i,j,k} (\partial_k H_{ij})^2$, and all vector norms are Euclidean. For periodic functions, we use K_p and cell averages. The identity $I_f''(t) = D(P_t f)$ will be proved for the density constructed here.

The proof reduces the sign to one radial integral, controls a smooth cutoff, and transfers the negative cell average to a probability density by a broad Gaussian envelope. The only finite computation is an integer interval evaluation on 8192 subintervals. The analytic estimates and the heat-flow argument are given below. Ordinary convexity is the target; log-convexity is a stronger property, discussed in [1].

2 The density and its radial kernel

2.1 A smooth periodic logarithmic bump

Definition 2.1 (The fixed density). *Let $d = 41$. Define $b(a) = \exp(-1/a)$ for $a > 0$ and $b(a) = 0$ for $a \leq 0$, and put*

$$\eta(v) = \frac{b(2-v)}{b(2-v) + b(v-1)}, \qquad u(s) = \tfrac{4}{3}\log(1 + e^{15 - 2s - s^2}), \tag{2.1}$$

$$\widetilde{u}(s) = \eta(s/16)u(s), \qquad w(x) = \widetilde{u}(|x|^2/2), \tag{2.2}$$

$$U(x) = \sum_{z \in \mathbb{Z}^{41}} w(x - 20z), \qquad q(x) = e^{U(x)}. \tag{2.3}$$

$$R = 10^{40}, \qquad Z = \int_{\mathbb{R}^{41}} q\gamma_{R^2}\, dx, \qquad f = Z^{-1} q\gamma_{R^2}. \tag{2.4}$$

The denominator defining η is positive everywhere. The function η is smooth, takes values in $[0, 1]$, equals 1 for $v \le 1$, and equals 0 for $v \ge 2$. The support of w lies in the closed ball of radius 8. Distinct lattice centers have distance at least 20, so these supports are disjoint. The sum defining U is locally finite and smooth. Thus $q \ge 1$ is periodic, and all its derivatives are bounded. In particular, $0 < Z < \infty$, and f is a smooth positive probability density with Gaussian decay. The width R is a standard deviation, so the envelope has variance R^2.

2.2 Radial tensor contractions

Lemma 2.2 (Radial reduction). *For $p(x) = e^{u(s)}$, with $s = |x|^2/2$, write $a = -u'(s)$, $B = u''(s)$ and $C = u'''(s)$. Then*

$$K_p(x) = K_d[u](s), \tag{2.5}$$

$$\begin{aligned} K_d[u](s) = {} & 8s^3C^2 + 24s^2BC + 6(d+2)sB^2 \\ & + 2\big((d-1)a^3 + (a - 2sB)^3\big). \end{aligned} \tag{2.6}$$

Proof. Differentiation gives $H_p = a\,\mathrm{Id} - Bxx^T$ and

$$\partial_{ijk} \log p = Cx_ix_jx_k + B(\delta_{ij}x_k + \delta_{ik}x_j + \delta_{jk}x_i). \tag{2.7}$$

The eigenvalues of H_p are a with multiplicity $d - 1$, and $a - 2sB$. At $x = re_1$, the nonzero third-derivative components are $Cr^3 + 3Br$ in position 111, and Br in the three permutations of $1jj$ for each $j > 1$. Squaring and summing gives $8s^3C^2 + 24s^2BC + 6(d+2)sB^2$. Rotational invariance and continuity at the origin complete the proof. □

For the profile in Definition 2.1, put $V = 15 - 2s - s^2$, $p_0 = e^V/(1 + e^V)$ and $V_s = -2 - 2s$. Its derivatives with respect to s are

$$u' = \tfrac{4}{3} p_0 V_s, \qquad u'' = \tfrac{4}{3}\big(-2p_0 + p_0(1 - p_0)V_s^2\big), \tag{2.8}$$

$$u''' = \tfrac{4}{3}\big(-6p_0(1 - p_0)V_s + p_0(1 - p_0)(1 - 2p_0)V_s^3\big). \tag{2.9}$$

The radial integral before introducing the cutoff is

$$F(r) = r^{40}(1 + e^{15 - r^2 - r^4/4})^{4/3} K_{41}[u](r^2/2), \qquad J = \int_0^8 F(r)\, dr. \tag{2.10}$$

The function F is smooth on a neighborhood of $[0, 8]$. In particular, r^{40} introduces no endpoint singularity.

3 A certified negative radial integral

Proposition 3.1 (Finite interval certificate). *The integral above satisfies*

$$-7710597700495949136 \;\le\; J \;\le\; -7375767767717917791 < -7 \cdot 10^{18}. \tag{3.1}$$

The certificate uses integer interval arithmetic at scale $S = 2^{160}$, followed by a midpoint remainder estimate. We give the enclosure rules and the resulting finite bounds.

3.1 Integer intervals and elementary functions

The pair (l, h) represents $[l/S, h/S]$. Rational constants are rounded outward. Addition and subtraction are exact at this scale. For multiplication, take the minimum and maximum of the four endpoint products, divide by S, and round outward. For a positive interval,

$$\left[\tfrac{l}{S}, \tfrac{h}{S}\right]^{-1} \subseteq \left[\tfrac{\lfloor S^2/h \rfloor}{S}, \tfrac{\lceil S^2/l \rceil}{S}\right]. \tag{3.2}$$

Negative intervals are handled by reflection. Division is used only when the denominator excludes zero. Odd integer powers are monotone; even powers use a separate zero-crossing case.

For an exponential endpoint x/S, choose $k \geq 0$ so that $z = |x|/(S2^k) \leq 1/8$. Enclose the Taylor polynomial through degree 32 by interval operations and use

$$0 \leq e^z - \sum_{j=0}^{32} \tfrac{z^j}{j!} \leq \tfrac{2}{8^{33}33!} < \tfrac{1}{S}. \tag{3.3}$$

The tail is bounded by a geometric series; the final inequality is the integer comparison $2S < 8^{33}33!$. Add one unit to the upper endpoint. For a negative original endpoint, take the reciprocal, and then square k times. Monotonicity gives the enclosure for a general input interval.

For a nonnegative endpoint l/S, compute an integer n with

$$n^3 \leq lS^2 < (n+1)^3. \tag{3.4}$$

This bounds its cube root between n/S and $(n+1)/S$. Evaluate $(1 + e^V)^{4/3}$ as $y\sqrt[3]{y}$, where $y = 1 + e^V \geq 1$. These operations all preserve containment, so their composition encloses the stated real expression.

3.2 Derivative enclosures and quadrature

A jet stores $(g, g', g''/2)$, with derivatives taken with respect to r. Each component encloses its value at every point of the input cell. Product coefficients are obtained by truncated polynomial multiplication. For powers, the implementation uses the usual first and second chain rules with interval powers; this retains the nonnegativity of squares.

$$(e^g)''/2 = e^g\left(g''/2 + (g')^2/2\right). \tag{3.5}$$

$$z = \sqrt[3]{g} > 0: \qquad z' = \tfrac{g'}{3z^2}, \qquad z''/2 = \tfrac{g''/2}{3z^2} - \tfrac{(g')^2}{9z^5}. \tag{3.6}$$

The value component of the logistic function is enclosed monotonically and intersected with $[0, 1]$. Its derivative components use the identities

$$p' = p(1-p)V', \qquad p''/2 = p(1-p)V''/2 + p(1-p)(1-2p)(V')^2/2. \tag{3.7}$$

Here V' and V'' are derivatives with respect to r, whereas the formulas for u in Section 2.2 use s derivatives. The program composes $r \mapsto s \mapsto V$ explicitly. Induction through the jet operations encloses $F''/2$ on the full cell.

Let $n = 8192$, $h = 8/n$, $m_j = (j + 1/2)h$. Suppose the computed midpoint enclosure is $[L_j, U_j]/S$ and the bound for $|F''/2|$ on the jth cell is B_j/S. Taylor expansion about the midpoint, with the odd linear term integrating to zero, yields

$$\tfrac{h}{S}\sum_j L_j - \tfrac{h^3}{12S}\sum_j B_j \leq J \leq \tfrac{h}{S}\sum_j U_j + \tfrac{h^3}{12S}\sum_j B_j. \tag{3.8}$$

Quantity	Outward integer bound
Midpoint sum, lower	-7543182734106933464
Midpoint sum, upper	-7543182734106933463
Absolute remainder, upper	167414966389015672
Integral, lower	-7710597700495949136
Integral, upper	-7375767767717917791

The table contains outward-rounded bounds from the finite integer calculation. Substitution into the midpoint inequality proves Proposition 3.1. The remainder bounds the real integral on each entire cell; it is not an estimate based only on sampled function values.

□

4 Cutoff and periodic averages

4.1 The cutoff error

Lemma 4.1 (Preservation of the radial sign). *For the cut profile in Definition 2.1, let*

$$J_{\mathrm{cut}} = \int_0^8 r^{40} e^{\widetilde{u}(r^2/2)} K_{41}[\widetilde{u}](r^2/2)\, dr. \tag{4.1}$$

$$|J_{\mathrm{cut}} - J| < 10^{-140}, \qquad J_{\mathrm{cut}} < -6 \cdot 10^{18}. \tag{4.2}$$

Proof. For $a > 0$, put $y = 1/a$. The first three derivatives of b are

$$b' = y^2 e^{-y}, \quad b'' = (y^4 - 2y^3)e^{-y}, \quad b''' = (y^6 - 6y^5 + 6y^4)e^{-y}. \tag{4.3}$$

Since $y^k e^{-y} \le k!$, their absolute values are at most 2, 36, and 1584. Every higher derivative is a polynomial in y times e^{-y}, so all derivatives tend to zero as $a \downarrow 0$. This proves smoothness of the zero extension.

On $[1, 2]$, write $A = b(2 - v)$ and $G = b(2 - v) + b(v - 1)$. One argument is at least $1/2$, hence $G \ge e^{-2} > 1/8$. Also $|A| \le 1$, and the derivative bounds for G are 4, 72, 3168. The reciprocal derivative formulas give bounds 8, 256, 20992, 2660352 for G^{-1} through order three. Applying the product rule to $\eta = AG^{-1}$ gives

$$\|\eta'\|_\infty < 300, \qquad \|\eta''\|_\infty < 30000, \qquad \|\eta'''\|_\infty < 3000000. \tag{4.4}$$

Consequently $\chi(s) = \eta(s/16)$ has derivative bounds 20, 120, 750. On the transition interval $16 \le s \le 32$, put $\tau = e^{15 - 2s - s^2}$. Then $0 < \tau \le e^{-273} < 10^{-100}$ and $|V_s| \le 66$. The formulas in Section 2.2 and $\log(1 + \tau) \le \tau$ imply

$$|u| < 2\tau, \quad |u'| < 100\tau, \quad |u''| < 6000\tau, \quad |u'''| < 400000\tau. \tag{4.5}$$

For the cut function, apply all product-rule terms. In particular,

$$\widetilde{u}''' = \chi u''' + 3\chi' u'' + 3\chi'' u' + \chi''' u. \tag{4.6}$$

Each of the first three derivatives of both profiles has absolute value at most $M = 10^6 \tau < 1$. In the radial kernel this yields

$$|K_{41}| \le 294976M^2 + 549330M^3 < 10^6 M^2 = 10^{18}\tau^2. \tag{4.7}$$

The two exponential weights are below 2. They agree for $s \le 16$. Bounding the integral of r^{40} by 8^{41}, we obtain

$$|J_{\mathrm{cut}} - J| < 4 \cdot 10^{18} \cdot 8^{41} \cdot 10^{-200} < 10^{-140}. \tag{4.8}$$

The last comparison follows from $e > 5/2$, $(5/2)^{273} > 10^{100}$ and $8^{41} < 10^{38}$. Proposition 3.1 now gives the second assertion.

□

4.2 The cell average and uniform bounds

Lemma 4.2 (Periodic estimates). *For $\Omega = [-10, 10]^{41}$, set $\langle g \rangle = 20^{-41} \int_\Omega g$ and $r_0 = 10^{10}/20^{41}$. Then*

$$\langle qK_q \rangle < -6r_0, \qquad 1 \le \langle q \rangle < 1 + 10^{-30}. \tag{4.9}$$

The four periodic functions q, qK_q, $q\mathrm{tr}(H_q^2)$ and $q\mathrm{tr}(H_q)$ each have absolute value below 10^{28}.

Proof. The area of the unit sphere satisfies

$$\omega_{40} = \tfrac{2^{41} 20!\, \pi^{20}}{40!} > 10^{-8}. \tag{4.10}$$

Outside the radius-8 ball in this cell, $q = 1$ and $K_q = 0$. Polar integration and Section 4.1 give

$$\langle qK_q \rangle = \tfrac{\omega_{40}}{20^{41}} J_{\mathrm{cut}} < -6r_0, \qquad r_0 = \tfrac{10^{10}}{20^{41}}. \tag{4.11}$$

Bounds for four periodic functions.

We need bounds for q, qK_q, $q\mathrm{tr}(H_q^2)$ and $q\mathrm{tr}(H_q)$. First $0 \le \widetilde{u} \le u < 22$, so $1 \le q < e^{22} < 10^{10}$. On the inner region, use $0 \le p_0 \le 1$ and $p_0(1 - p_0) \le 1/4$ in Section 2.2; on the transition region, use Section 4.1. Everywhere the profile is nonconstant,

$$|\widetilde{u}'| < 100, \quad |\widetilde{u}''| < 1500, \quad |\widetilde{u}'''| < 10^5, \quad 0 \le s \le 32. \tag{4.12}$$

The radial Hessian eigenvalue therefore has absolute value below $100 + 64 \cdot 1500 = 96100$, while each tangential eigenvalue has absolute value below 100. Consequently

$$|K_q| < 4400152418000000 < 10^{16}, \tag{4.13}$$

$$|\mathrm{tr}(H_q^2)| < 40 \cdot 100^2 + 96100^2, \qquad |\mathrm{tr}(H_q)| < 40 \cdot 100 + 96100. \tag{4.14}$$

Multiplication by $q < 10^{10}$ shows that all four functions have absolute value below 10^{28}. The trace in the last expression may have either sign.

Average mass.

For $a \ge 0$, integrating $(1 + a)^{1/3} \le 1 + a^{1/3}$ gives $(1 + a)^{4/3} - 1 \le (4/3)(a + a^{4/3})$. Since cutting off decreases the weight, dropping the quartic term in the exponent and using Gaussian integrals gives

$$\int_{\mathbb{R}^{41}} (e^{w(x)} - 1)\, dx \le \tfrac{4}{3}\big(e^{15}\pi^{41/2} + e^{20}(3\pi/4)^{41/2}\big). \tag{4.15}$$

Using $\pi < 4$ and $e < 11/4$, division by 20^{41} bounds the excess cell average by

$$\tfrac{8 \cdot 4^{20}}{3 \cdot 20^{41}}\big((11/4)^{15} + (11/4)^{20}\big) < 10^{-30}. \tag{4.16}$$

Thus $1 \le \langle q \rangle < 1 + 10^{-30}$. The sphere-area lower bound uses $\pi > 3$ and an exact rational comparison.

□

5 Gaussian localization and heat flow

5.1 Gaussian averages of periodic functions

Lemma 5.1 (Gaussian averaging). *Let g be continuous and L-periodic in each coordinate, with $|g| \le M$, and let $\langle g \rangle$ denote its average over a period cell. Let G_R be centered Gaussian with covariance $R^2\mathrm{Id}$, and set $A = 2\pi^2 R^2/L^2$. Then*

$$\begin{aligned} |\mathbb{E}g(G_R) - \langle g \rangle| &\le M \sum_{m \in \mathbb{Z}^d \setminus \{0\}} e^{-A|m|^2} \\ &\le M\Big(\Big(1 + \tfrac{2e^{-A}}{1-e^{-3A}}\Big)^d - 1\Big). \end{aligned} \tag{5.1}$$

Proof. Periodize the Gaussian: $W_R(x) = \sum_z \gamma_{R^2}(x + Lz)$. Gaussian decay implies uniform convergence on the cell. Integrating over translated cells gives its Fourier coefficients $L^{-d}e^{-A|m|^2}$. The associated Fourier series converges absolutely and uniformly and equals W_R, by uniqueness of Fourier coefficients for continuous periodic functions. Integrate gW_R on the cell and bound the nonzero modes by M. For the last inequality, use $k^2 \ge 1 + 3(k-1)$ for integers $k \ge 1$ and then factor the lattice sum by coordinates. This works for signed g. □

For $d = 41$, $L = 20$, $R = 10^{40}$ and $M = 10^{28}$, we have $A > 1000$ and $z = e^{-A} < 10^{-300}$. Since $1/(1-z^3) < 2$ and $(251/250)^{40} < 2$,

$$(1+4z)^{41} - 1 \le 164z(1+4z)^{40} < 328z. \tag{5.2}$$

The averaging error is therefore less than 10^{-80} for each of the four functions from Lemma 4.2. In particular,

$$1 \le Z = \mathbb{E}q(G_R) < 1 + 10^{-30} + 10^{-80} < 2. \tag{5.3}$$

5.2 Negativity for the fixed density

Proposition 5.2 (The static inequality). *The density of Definition 2.1 satisfies $D(f) < -10^{-44}$.*

Proof. Since $H_f = H_q + R^{-2}\mathrm{Id}$ and $\nabla H_f = \nabla H_q$, expansion of the matrix cube gives

$$\begin{aligned} ZD(f) = {} & \mathbb{E}(qK_q)(G_R) + 6R^{-2}\mathbb{E}(q\,\mathrm{tr}(H_q^2))(G_R) \\ & + 6R^{-4}\mathbb{E}(q\,\mathrm{tr}(H_q))(G_R) + 82R^{-6}Z. \end{aligned} \tag{5.4}$$

The common absolute bound, rather than any assumption on the sign of the trace, now yields

$$ZD(f) < -6r_0 + 10^{-80} + 6 \cdot 10^{28}R^{-2} + 6 \cdot 10^{28}R^{-4} + 164R^{-6} < -5r_0. \tag{5.5}$$

The final strict comparison follows by substituting $R = 10^{40}$ and $r_0 = 10^{10}/20^{41}$. Because $0 < Z < 2$, dividing this negative upper bound gives

$$D(f) < -\tfrac{5}{2}r_0 < -10^{-44}. \tag{5.6}$$

□

5.3 Regularity and continuity at the initial density

Lemma 5.3 (Heat-flow regularity). *For the fixed density f, the Fisher information is twice continuously differentiable at positive times, the integrations by parts below are valid, and $D(P_t f) \to D(f)$ as $t \downarrow 0$.*

Proof. Set $T = R^2$ and define

$$\alpha_t = \tfrac{T}{T+t}, \qquad \beta_t = \tfrac{Tt}{T+t}, \qquad h_t = P_{\beta_t} q. \tag{5.7}$$

Completing the square in the product of Gaussian densities gives

$$\gamma_T(y)\gamma_t(x-y) = \gamma_{T+t}(x)\gamma_{\beta_t}(y-\alpha_t x). \tag{5.8}$$

Multiply by $q(y)$ and integrate to obtain the exact representation

$$P_t f(x) = Z^{-1}\gamma_{T+t}(x) h_t(\alpha_t x). \tag{5.9}$$

Every derivative of the periodic function q is bounded and uniformly continuous. Differentiation of q under a Gaussian expectation is justified by bounded domination. Hence, for any multi-index ν,

$$1 \le h_t \le \|q\|_\infty, \qquad \partial^\nu h_t = P_{\beta_t}(\partial^\nu q), \qquad \|\partial^\nu h_t\|_\infty \le \|\partial^\nu q\|_\infty. \tag{5.10}$$

The Gaussian approximate identity gives uniform convergence of each fixed spatial derivative of h_t to that of q as $t \downarrow 0$. Since $h_t \ge 1$, the same holds for any positive-order spatial derivative of $\log h_t$. Thus

$$H_{P_t f}(x) = (T+t)^{-1}\mathrm{Id} - \alpha_t^2(\nabla^2 \log h_t)(\alpha_t x) \tag{5.11}$$

and its spatial derivative are uniformly bounded for $0 \le t \le t_1$ and converge at every fixed x to their initial values. Pointwise convergence suffices here; $\alpha_t x \to x$ is not asserted uniformly in x.

For a constant depending on T, t_1, q, but not on t or x,

$$P_t f(x) \le C\exp\left(-\tfrac{|x|^2}{2(T+t_1)}\right) \qquad (0 \le t \le t_1). \tag{5.12}$$

Dominated convergence therefore implies $D(P_t f) \to D(f)$. The score $\nabla \log(P_t f)$ grows at most linearly, so Fisher information is finite as well.

Time derivatives and integration by parts.

The Gaussian heat equation and integration by parts against the bounded smooth periodic function give $\partial_t h_t = (\beta_t'/2) P_{\beta_t} \Delta q$. Repetition supplies all required finite time derivatives. Combining this with $\alpha_t x$ and the explicit Gaussian factor shows that differentiated integrands are bounded by a polynomial in $|x|$ times a fixed integrable Gaussian, on every compact time interval. This justifies differentiation under the integral. For spatial integration by parts, insert smooth compactly supported cutoffs with growing radius; polynomial-Gaussian domination makes the boundary terms tend to zero.

□

5.4 Derivative identities and completion of the proof

Write $\rho = P_t f$, $v = \log \rho$, $w = \nabla v$ and $A = \nabla^2 v$. Lemma 5.3 justifies the differentiations and integrations by parts. The heat equation gives

$$v_t = \tfrac{1}{2}(\Delta v + |w|^2), \qquad w_t = \tfrac{1}{2}\Delta w + Aw, \tag{5.13}$$

$$A_t = \tfrac{1}{2}\Delta A + w \cdot \nabla A + A^2. \tag{5.14}$$

The derivative of an expectation obeys $\frac{d}{dt}\int\rho\phi = \int\rho(\phi_t + \Delta\phi/2)$. Applying it to $\phi = |w|^2$ gives

$$I_f'(t) = \int\rho\left(2w\cdot\Delta w + |A|_F^2 + 2w\cdot Aw\right)dx. \tag{5.15}$$

Weighted integration by parts yields $\int\rho\, w\cdot\Delta w = -\int\rho|A|_F^2 - \int\rho\, w\cdot Aw$. Consequently

$$I_f'(t) = -Q(\rho), \qquad Q(\rho) = \int\rho|A|_F^2\,dx. \tag{5.16}$$

Differentiate Q in the same way:

$$\frac{d}{dt}Q(\rho) = \int\rho\left(2A:\Delta A + |\nabla A|_F^2 + 2A:(w\cdot\nabla A) + 2\mathrm{tr}(A^3)\right)dx. \tag{5.17}$$

The weighted integration by parts formula for $A:\Delta A$ cancels the transport term and one copy of the gradient square, leaving

$$\frac{d}{dt}Q(\rho) = \int\rho\left(-|\nabla A|_F^2 + 2\mathrm{tr}(A^3)\right)dx = -D(\rho). \tag{5.18}$$

The last sign uses $H_\rho = -A$. Thus these are identities for actual time derivatives:

$$I_f''(t) = D(P_t f) \qquad (t>0). \tag{5.19}$$

Proof of Theorem 1.1. Proposition 5.2 gives $D(f)<0$. Lemma 5.3 implies $D(P_t f)<0$ for every sufficiently small positive t. The derivative identity therefore gives $I_f''(t)<0$ on an interval $(0,\varepsilon)$. A twice differentiable convex function has nonnegative second derivative at every interior point, which proves Theorem 1.1.

□

6 Conclusion

The construction fixes a radial logarithmic bump, a smooth cutoff at radius 8, period 20, and a Gaussian envelope of standard deviation 10^{40}. Integer interval arithmetic gives a negative radial integral. The cutoff and Gaussian averaging errors are smaller than the negative margin, and the explicit Gaussian correction terms preserve it. Thus the resulting probability density satisfies

$$D(f) < -10^{-44}. \tag{6.1}$$

The Gaussian product identity supplies the regularity and domination needed to connect this static inequality to the heat flow. Continuity at the initial density and the two derivative identities give

$$I_f''(t) < 0 \qquad (0<t<\varepsilon). \tag{6.2}$$

This proves the asserted failure of ordinary convexity in dimension 41, subject to the finite interval calculation in Proposition 3.1. The computation was performed with integer arithmetic; its rules and aggregate bounds are recorded in Section 3. The full statement has not yet been formalized in Lean. No minimal-dimension assertion is made.

Reference

[1] J. Zou, L. Fan, J. Gao and J. Wang. A Hexagonal Counterexample to Log-Convexity of Fisher Information Along the Heat Flow. arXiv:2605.18081v3, 2026. [online]